Learn to Hack Hackathon

Group 1 – Matrix Not Found

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Problem Statement

Buildings are a major contribution to carbon emissions through energy usage, and we are to identify which need to be retrofitted with an **ML model** that **predicts building energy consumption** based off known factors, such as (but not limited to):

- Floor area in square feet
- Energy Star rating
- Average location temperature
- Precipitation and snowfall

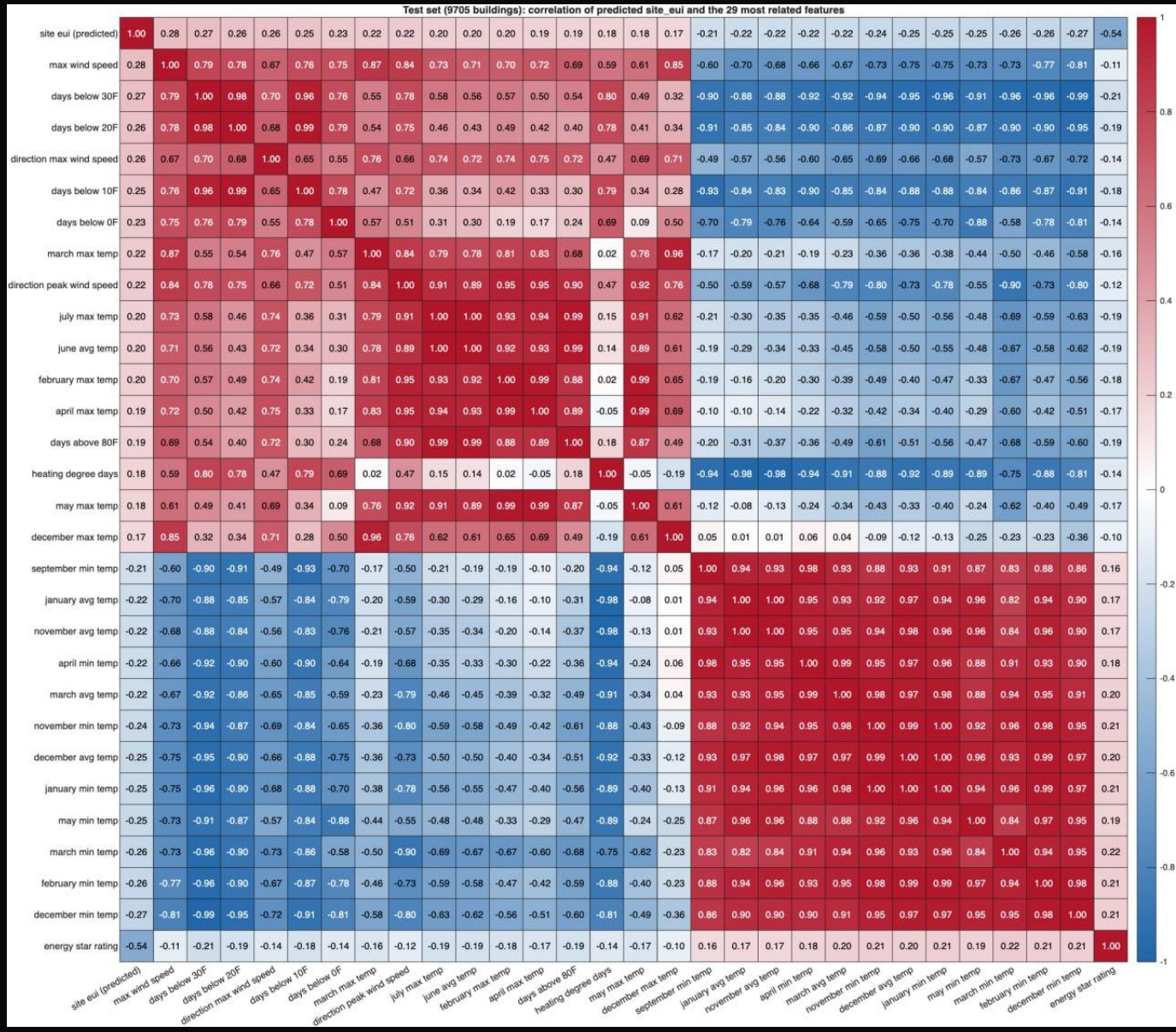
The model is meant to advise leaders on which buildings need to be improve, based off predicted energy usage.



Exploratory Data Analysis

To begin, we had to perform an EDA in order to take out incomplete entries, and determine which factors had strong correlations to Site Energy Usage Intensity (referred to as “site_eui” within given datasets).

From initial observation, the “energy star rating” factor has the **strongest correlation** to site_eui, with a magnitude of **0.54**.



Model Architecture

Tree models were used for regression.

LSBoost v1

- All possible factors from dataset used

LSBoost v2

- Selected factors based off heatmap

LSBoost v3

- Same as previous, with AI-selected features.

Results

Five-fold pooled out-of-fold

Parameter	LSBoost v1	LSBoost v2	LSBoost v3
MAE	23.3941	23.3385	21.0390
MSE	2044.3723	2024.7375	1695.0047
RMSE	45.2147	44.9971	41.1704
R^2	0.3976	0.4034	0.5005

Latest-year holdout metrics

Parameter	LSBoost v1	LSBoost v2	LSBoost v3
MAE	22.7927	22.7324	21.0440
MSE	2023.2171	2003.8279	1842.3056
RMSE	44.9802	44.7641	412.9221
R^2	0.3777	0.3826	0.4333

Conclusion

All of our models were successful in predicting energy usage, but our best by a far margin is **LSBoost v3**. This is based off resultant model parameters.

In real-world use cases, we believe that **Model v3** will be of use in

Repository link:

Roh00t/MatrixNotFound: LearnToHack